- **Hierarchical**—Namespaces are often structured hierarchically. The Domain Name System (DNS) is a well-known example of a hierarchical namespace, and that structure is essential to the way domain names work. Other notable examples of hierarchical namespaces include X.509 certificates and Lightweight Directory Access Protocol (LDAP). Hierarchical namespaces facilitate discovery because the name itself tells you how to look it up, so long as you know the root. And each node can be granted local control over the names beneath it without risking collisions.
- **Flat**—Flat namespaces have no structure. The chief advantage of flat namespaces is that names can be used across any context without risk of collision. There are several ways to do this. In the case of UUIDs, collisions are avoided algorithmically. Blockchains or other decentralized data stores can also be used.
- **Federated**—A federated namespace is a distributed naming system that works by sharing the responsibility of assigning names to several organizations or services. The primary advantage of federated namespaces is local control with global discovery. Email is probably the best example of a federated namespace. Each email provider is responsible for assigning unique names within its domain. Federated namespaces require governance so that participants in the federation know how to discover the meaning of names managed by other participants.

The choice of namespace is strategic, influencing not only how names are managed internally but also how easily identities can move between systems, how secure naming is against impersonation, and how easily identities can be discovered when needed for authorization. A poorly chosen namespace locks identities into silos, increases security risks, and complicates policy enforcement. In the next section, we'll explore how the model for an identity system supports the right namespace.

2.5 Identity system models

Naming and discovery are closely tied to the model of the identity system itself. A namespace doesn't exist in isolation—it reflects the underlying architecture of the identity system that manages it. Whether an organization